

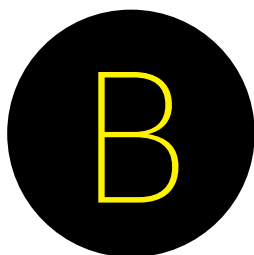


BY RAHUL MUNJAL

Renewables will Power the New India

As India transforms into an economic powerhouse, it needs to plan well and address the challenges around green energy

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BEING HOME TO almost 18 per cent of the world's population, India's economic growth is of global interest. Latest studies indicate that a young workforce, strong manufacturing sector and rapidly growing digital economy will enable India to witness a sustained GDP growth of 7.5-8 per cent. India's consumption levels are poised for exponential growth as well with per capita income increasing sixfold. Moreover, as the world gears up to combat the looming threat of climate change, all eyes are on how a rapidly growing India develops its energy sector. The size of India's grid is expected to expand to almost 2,300 GW by 2050 as per capita consumption of electricity catches up with the world's average. Maintaining this growth

trajectory, while addressing energy security will depend on affordable and reliable clean energy. And the signs are extremely encouraging. India's wind and solar capacity has quadrupled to 100 GW in a decade, and the cost of power generated from wind and solar is amongst the lowest in the world, thereby making India one of the most attractive destinations for clean energy investment.

I envision that by 2047, India will have universal power access with renewable energy as the dominant source of power. I wish to see healthy competition between electric utilities and consumers enjoying the freedom of choosing their service provider. Electric vehicles (EVs) should be ubiquitous, enabling cities to breathe cleaner air. India would very likely be an innovation hub in low-carbon technologies and a global green manufacturing player with high output mega factories. In particular, I would like to see India as the green H2 capital of the world and amongst its lowest cost producers. I also hope to see steel, cement, and fertiliser as certified green products and a circular economy reducing wastage and increasing competitiveness of our industries.

For this vision to be translated into reality, we need to plan well and address certain challenges. Firstly, India would have to add around 20-30 GW of renewable energy capacity per annum. All this wind and solar energy must be rapidly integrated into the grid. Misalignment between demand and supply peaks must be managed.

Secondly, achieving supremacy in next-generation clean technologies require investments in innovation, R&D and infrastructure. Incentivising investments in development of such risky technologies in an uncertain future and limited or non-existent ecosystem is a challenge. Next, the poor financial health of state-owned discoms has dampened investor sentiment and can decelerate sectoral growth. This must be addressed.

Lastly, India's land acquisition process for its renewable projects is slow and tedious. The reasons being small land holdings, poor land records, and state-specific procedures. Additionally, the choice of land is restricted by the availability of power evacuation capacity nearby.

Several measures must be taken by policymakers and private players to achieve system flexibility for